
CF00: COMPLETE FORMALISM - INDUCED GEOMETRY AND EMERGENT DYNAMICS IN VDM

COMPLETE FORMALISM

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ABSTRACT

Beginning from the post-2D invariant stack inherited from CF000 — which delivers proto- $\mathbb{N}$, the algebra unlock, and the imaginary unit as ORS expressed algebraically — this paper first derives the real completion, topology, the primitive half-turn constant π of the continuous ORS orbit, the carrier domain $\mathcal{M}$, the effective complex Hilbert bundle, and local $U(1)$ redundancy, and then derives quotient-physical local variation, the quantum geometric tensor, the induced metric and curvature sectors, the mixed reversible/irreversible dynamical architecture, constitutive narrowing, and locality-bearing support without inserting primitive adjacency, support, bond, gauge, or engine structure by hand. In this way CF00 shows how induced geometry and emergent dynamics arise on the derived carrier, stops treating π as an imported constant, and prepares the explicit QGT-to-metriplectic bracket mapping taken up in CF01.

Keywords post-2D invariant stack · primitive half-turn constant · ORS orbit · carrier derivation · quantum geometric tensor · induced geometry · emergent dynamics · VDM

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1. Executive Summary

This paper is the first carrier-stage bridge between CF000 and CF01 in the VDM program. Beginning from the post-2D invariant stack inherited from CF000 — which delivers proto- $\mathbb{N}$, the algebra unlock, and the imaginary unit as ORS expressed algebraically — this paper first derives the real completion, topology, the primitive half-turn constant π of the continuous ORS orbit, the carrier domain $\mathcal{M}$, the effective complex Hilbert bundle, and local $U(1)$ redundancy, and then derives quotient-physical local variation, the quantum geometric tensor, the induced metric and curvature sectors, the mixed reversible/irreversible dynamical architecture, constitutive narrowing, and locality-bearing support without inserting primitive adjacency, support, bond, gauge, or engine structure by hand.

CF00 does not reopen the pre-carrier origin burden. That burden belongs to CF000. The stage-correct question here is: given the inherited post-2D invariant stack, what is the first lawful host of continued invariant-bearing articulation, and what quotient-physical geometric and dynamical structures are forced once representative variation on that derived carrier is quotiented by derived $U(1)$ redundancy?

The answer carried here is that the inherited stack forces, in order, the real completion needed for carrier coordinates, the complex field licensed by the 2D ORS algebra, the topological and C^1 variation structure needed for lawful local variation, the continuous completion of the ORS orbit, the primitive half-turn constant π , the carrier domain $\mathcal{M}$ as first admissible host, the effective complex Hilbert bundle over that domain, and local $U(1)$ redundancy as the natural phase redundancy of normalized complex fibers. Once those objects are earned, quotient-physical variation on $\mathcal{M}$ forces the QGT, the induced metric and curvature sectors, the mixed reversible/irreversible architecture, constitutive narrowing, and derived support. In this way CF00 serves as the rigorous bridge from the post-2D invariant closure of CF000 to the explicit QGT-to-metriplectic program taken up in CF01 while locating π within the canon as a derived invariant constant rather than an imported mathematical datum.

2. Root Status, Inherited Stack, and Notation Convention

2.1 Stage status

CF00 is the first carrier-stage Complete Formalism after CF000. CF000 closes the pre-carrier route through 2D and hands forward not a primitive carrier, but the post-2D invariant stack: the primitive bifurcation invariant, the 1D axis-proper effective invariant (continuous same-axis bifurcation, proto- $\mathbb{N}$), and the 2D axis-proper effective invariant (the numerical relation-grid invariant together with algebra unlock). CF00 therefore functions as the first stage at which that inherited stack is used to derive the carrier domain and then the induced geometry and emergent dynamics built on that derived carrier.

Any downstream document in this branch that uses quotient-physical variation, induced geometry, scalar-generated mixed dynamics, constitutive reservoir closure, or support-derived locality inherits those structures from CF00.

2.2 Notation convention

Two different kinds of object must not share the same symbol.

- $\mathcal{M}$ denotes the carrier domain derived at the CF00 stage: the arena on which the representative state-family is defined.
- M denotes the dissipative symmetric operator or sector in the emergent dynamical split $J \oplus M$.

This distinction is ontological, not cosmetic.

- $\mathcal{M}$ is a carrier arena.
- M is an emergent operator on quotient cotangent data.

No statement in this document uses plain M for the carrier domain. All carrier-domain quantities use $\mathcal{M}$ and derived notation such as $d_{\mathcal{M}}$ for the carrier-domain distance.

2.3 The stage question

The stage question for CF00 is:

Given the post-2D invariant stack inherited from CF000, what first lawful carrier/domain structure is forced, and what quotient-physical geometric and dynamical structures are then forced once local normalized representative variation is quotiented by derived $U(1)$ redundancy?

The answer of CF00 is:

The inherited stack is sufficient to derive the carrier domain $\mathcal{M}$, the effective complex Hilbert bundle over $\mathcal{M}$, and local $U(1)$ redundancy; and those derived inputs are sufficient to derive quotient-physical local variation, the quantum geometric tensor, the induced metric and curvature sectors, the mixed reversible/irreversible architecture, constitutive narrowing, and locality-bearing support without any additional primitive graph, support, gauge, or engine structure.

2.4 Inherited stack and stage scope

Theorem 2.4.1 (Inherited invariant stack and derivation scope).

Within CF00, the post-2D invariant stack delivered by CF000 is the inherited primitive input of this stage. That stack consists of: the primitive bifurcation invariant (unresolved two-pole opposition), the 1D axis-proper effective invariant (continuous same-axis bifurcation, proto- $\mathbb{N}$), and the 2D axis-proper effective invariant (numerical relation-grid invariant, algebra unlock). CF00 does not reopen the pre-carrier origin burden. Its formal burden is to derive, from that stack, the first lawful host/carrier structure and then the quotient-physical geometric and dynamical structures forced by representative variation on that derived carrier.

Status. Proved in this CF.

Proof. CF000 closes the $0D \rightarrow 1D \rightarrow 2D$ route and delivers: proto- $\mathbb{N}$ at 1D, ordered-pair algebra and $\mathbb{C}$ (as ORS expressed algebraically) at 2D, and the explicit handoff requirement that CF00 must derive $\mathcal{M}$ rather than assume it. CF00 therefore begins from the inherited stack as given input and derives, in strict order:

1. $\mathbb{R}$ from proto- $\mathbb{N}$ via the standard algebraic completion chain;
2. $\mathbb{C}$ from the 2D ORS rotation operator already licensed by CF000 §7.4.1;
3. topology on the completed number structure;
4. the continuous completion of the discrete ORS quarter-turn on the two-axis carrier implied by the real completion and the topological structure;
5. the primitive half-turn constant π as the unique least positive opposite-pole attainment of that continuous ORS completion;
6. differentiability as lawful variation on the topological structure;
7. the carrier domain $\mathcal{M}$ as the first admissible host of continued invariant-bearing articulation past 2D;
8. the effective Hilbert bundle over $\mathcal{M}$ using $\mathbb{C}$ from step 2;
9. local $U(1)$ redundancy from the complex phase structure acting on normalized fibers.

Only after step 7 does the representative ontology of §3 become valid. The formal burden of the pre-carrier origin belongs to CF000. $\square$

2.5 Derivation of the carrier domain from the inherited invariant stack

2.5.1 From proto- $\mathbb{N}$ to $\mathbb{R}$

Theorem 2.5.1 ($\mathbb{R}$ as forced completion of proto- $\mathbb{N}$).

The proto-natural-number structure licensed at 1D by CF000 generates, under successive forced completions, the real number field $\mathbb{R}$.

Status. Proved in this CF.

Proof. Four completion steps are forced by the same saturation logic that forced $1D \rightarrow 2D$ in CF000.

- $\mathbb{N} \rightarrow \mathbb{Z}$: Lawful binary combination, licensed at 2D, must be closed under the inverse of addition. If $a - b$ is not always admitted, the set is not saturated with respect to lawful combination. Therefore the integers $\mathbb{Z}$ are forced as the closure of proto- $\mathbb{N}$ under additive inversion.
- $\mathbb{Z} \rightarrow \mathbb{Q}$: Multiplication is licensed as the second lawful binary combination on the two-axis ordered-pair structure (Theorem 4.9.1 of CF000). Its inverse, division, is then forced by the same closure logic. Closure under multiplicative inversion, where the divisor is nonzero, forces $\mathbb{Q}$.

- $\mathbb{Q} \rightarrow \mathbb{R}$: The rationals contain Cauchy sequences that do not converge within $\mathbb{Q}$. A system that admits lawful sequential structure but contains non-terminating sequences without limits is not same-axis saturated: those limit points are lawful articulations that remain unborne. Cauchy completion therefore forces $\mathbb{R}$.

Thus the real number field $\mathbb{R}$ is the first complete ordered field derivable from the CF000 handoff. No external postulate is required. $\square$

2.5.2 $\mathbb{C}$ as the 2D ORS algebra

Theorem 2.5.2 ($\mathbb{C}$ as licensed by CF000).

The complex number field $\mathbb{C} = \mathbb{R} \oplus i\mathbb{R}$ is the natural algebra of the two-axis invariant-bearing structure admitted at 2D. The imaginary unit i is the algebraic form of the orthogonal rotation operator ORS, satisfying $i^2 = -1$ because returning along the orthogonal rotation path reaches the opposite pole of the invariant.

Status. Proved in this CF.

Proof. CF000 §7.4.1 proves this directly. The rotation operator R satisfying $R(1, 0) = (0, 1)$ and $R(0, 1) = (-1, 0)$ has $R^2 = -1$ because the two poles are irreconcilable. This is exactly i . The field $\mathbb{C} = \mathbb{R}[i]/(i^2 + 1)$ is therefore inherited from the CF000 closure, not postulated. $\square$

2.5.3 Topology from ordered completion

Definition/Theorem 2.5.3 (Topology forced by ordered completion).

The ordering structure of $\mathbb{R}$ together with the completion metric $d(a, b) = |a - b|$ generates a natural topology on $\mathbb{R}$ via the open-ball neighborhood structure. The product topology on $\mathbb{R}^n$ for finite n follows. This is not an external import: it is the lawful closure/completion burden expressed as topological structure.

Status. Derived structural consequence.

Proof. Once the real completion is present, open neighborhoods are fixed by the induced metric and ordered-field compatibility. No additional primitive topological axiom is required to specify which local neighborhoods count as admissible. The topology is the one already determined by ordered completion and the completion metric. $\square$

2.5.3b π as the primitive half-turn constant of the ORS orbit

Theorem 2.5.3b.1 (Continuous completion of ORS).

Once the 2D ORS quarter-turn of CF000, the real completion of Theorem 2.5.1, and the topological continuity licensed by ordered completion are in hand, the discrete ORS action admits a continuous one-parameter completion $\{R_\theta\}_{\theta \in \mathbb{R}}$ on the two-axis carrier, satisfying

$$R_0 = I, \quad R_{\theta+\phi} = R_\theta R_\phi,$$

and extending the already-earned discrete quarter-turn action.

Status. Derived structural consequence.

Proof. CF000 closes the discrete orthogonal re-articulation step and yields the quarter-turn operator whose square is the opposite-pole map:

$$R^2 = -I.$$

Theorem 2.5.1 supplies the completed ordered field needed for continuous parameterization, and Theorem 2.5.3 supplies the topological continuity structure determined by that completion. Once those are in hand, the discrete ORS action admits a lawful continuous hull: a one-parameter family of norm-preserving rotations on the two-axis carrier. No new primitive group object is inserted here. This is the continuous completion of an already-earned discrete action. $\square$

Definition 2.5.3b.2 (Primitive half-turn constant).

Let $\{R_\theta\}_{\theta \in \mathbb{R}}$ be the continuous ORS completion on the two-axis carrier. Define

$$\pi := \inf\{\theta > 0 : R_\theta = -I\}.$$

This is the first positive rotation parameter at which the invariant reaches its opposite pole. It is the minimal angular cost of maximal irreconcilability.

Theorem 2.5.3b.3 (π as the unique least positive half-turn).

The quantity π defined above is the unique least positive parameter at which the ORS orbit reaches the opposite pole. Hence π is not primitively imported from circle-area, trigonometric series, or external convention. It is first earned as the half-turn constant of the invariant's own continuous rotational expression.

Status. Derived structural consequence.

Proof. Because the discrete ORS seed satisfies $R^2 = -I$, two successive quarter-turn applications reach the opposite pole. Therefore the set

$$H := \{\theta > 0 : R_\theta = -I\}$$

is nonempty. Also $R_0 = I \neq -I$. By continuity of $\theta \mapsto R_\theta$ at 0, there exists $\varepsilon > 0$ such that $R_\theta \neq -I$ for every $0 < \theta < \varepsilon$. Hence

$$H \subseteq [\varepsilon, \infty),$$

so the opposite-pole hit set is bounded away from 0. By continuity of the completed orbit, H is closed in $(0, \infty)$. Since H is nonempty and bounded below by the positive number ε , order-completeness gives an infimum $\inf H \in [\varepsilon, \infty)$, and closedness ensures that this infimum belongs to H . Thus there exists a least positive parameter at which the orbit reaches the opposite pole. By leastness, that parameter is unique. This is exactly the quantity defined above as π . $\square$

Corollary 2.5.3b.4 (Geometric realization of the half-turn constant).

Under the continuous ORS completion, the orbit of a unit carrier point is the unit circle, the region it encloses is the unit disk, and the geometric measure constant attached to that enclosed disk is the same constant π already defined above. In particular,

$$\pi = \iint_{D^2} dx dy, \quad D^2 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$

Status. Derived geometric corollary.

Proof. Once the ORS action is continuously completed on the two-axis carrier, the orbit of a unit carrier point closes into the unit circle. The enclosed region is the unit disk. The geometric measure constant attached to that disk is not a new primitive definition; it is the geometric realization of the already-earned half-turn constant. $\square$

Remark. This subsection does not yet claim transcendence. Its completed claim is narrower and structurally earlier: CF00 earns π as the primitive half-turn constant of the continuous ORS orbit, and only then recovers unit-circle and unit-disk expressions as downstream consequences. The transcendence branch, the identity $e^{i\pi} = -1$, chirality as oriented half-turn boundary, and the Nielsen–Ninomiya reinterpretation remain the work of CF13.

2.5.4 Differentiability from lawful variation

Definition/Theorem 2.5.4 (C^1 differentiability as forced variation structure).

On a topological carrier equipped with the real number field, a local variation structure is a map that assigns to each point a linear approximation of the change in a field value under small displacement. Requiring that this approximation be well-defined and consistent — that the error in the linear approximation goes to zero faster than the displacement — is exactly C^1 differentiability.

Status. Derived structural consequence.

Proof. The invariant cannot discharge, so local variation cannot be arbitrary if continued articulation is to remain lawfully comparable from point to point. The weakest regularity that keeps local variation well-defined, linearly comparable, and stable under the quotient-physical constructions that follow is C^1 . Anything weaker fails to support a consistent projected-variation calculus; anything stronger is not yet forced at the minimum stage of CF00. For the downstream QGT-to-metriplectic bracket program of CF01, however, the projected variation constructions of §4–5 must be iterated to higher order. The same forcing argument that requires C^1 — invariant-bearing local variation must be linearly comparable — applies recursively: second-order variation must be linearly comparable to first-order variation, and so on. This forces C^∞ smoothness as the saturation of the differentiability burden in the presence of iterated variation. CF00 operates at C^1 as the minimum; CF01 may stipulate C^∞ as the iterated completion of that minimum, which is licensed by the same invariant logic applied recursively. $\square$

2.5.5 The carrier domain $\mathcal{M}$ as derived host

Theorem 2.5.5 ($\mathcal{M}$ as the first admissible host of continued invariant-bearing articulation past 2D).

Given $\mathbb{R}$ (Theorem 2.5.1), $\mathbb{C}$ (Theorem 2.5.2), topology (Theorem 2.5.3), the primitive half-turn constant π (Theorem 2.5.3b.3), and C^1 differentiability (Theorem 2.5.4), the carrier domain $\mathcal{M}$ is defined as a connected C^1 domain admitting local coordinates in $\mathbb{R}^n$ for some finite n .

$\mathcal{M}$ is not assumed. It is the first structure that can host continued invariant-bearing articulation past the 2D algebraic stage: it has a real number field for coordinates, a complex field for the representative state fiber, a topology for

local variation to be well-defined, the primitive half-turn constant π already fixed by the continuous ORS orbit, and differentiability for the projected variation construction of §4 to proceed.

Status. Proved in this CF.

Proof. Nothing weaker than a connected C^1 coordinate domain can host the local representative variation, local continuity, and projected derivative constructions used later in the paper. Nothing richer is yet forced at stage entry. The first admissible host is therefore the connected C^1 carrier domain $\mathcal{M}$ built on the already earned number, topological, and variation structure. It is primitive input for derivations in §3 onward only in the sense that it is now earned before being used. $\square$

2.5.6 Hilbert bundle from $\mathcal{M}$ and $\mathbb{C}$

Theorem 2.5.6 (Effective Hilbert bundle as derived structure).

Given $\mathcal{M}$ (Theorem 2.5.5) and $\mathbb{C}$ (Theorem 2.5.2), the effective complex Hilbert bundle $\pi : \mathcal{H} \rightarrow \mathcal{M}$ is the minimal fiber bundle over $\mathcal{M}$ with fibers carrying inner-product structure over $\mathbb{C}$ sufficient to define normalization, overlap, projected variation, and local phase redundancy.

Status. Proved in this CF.

Proof. The quotient-physical constructions of §4 require a normalized representative state at each point of the carrier together with a fiberwise inner product that can distinguish self-parallel from horizontal variation. Once $\mathcal{M}$ and $\mathbb{C}$ are available, the minimal structure that supports those operations is an effective complex Hilbert bundle. The bundle is therefore forced by what later derivations require; it is not a new primitive insertion. $\square$

2.5.7 $U(1)$ redundancy from $\mathbb{C}$ phase structure

Theorem 2.5.7 (Local $U(1)$ redundancy as derived from $\mathbb{C}$).

Given the complex Hilbert bundle (Theorem 2.5.6) and normalized fibers, the local phase redundancy $|\psi(x)\rangle \sim e^{i\Lambda(x)}|\psi(x)\rangle$ with $\Lambda \in C^1(\mathcal{M}, \mathbb{R})$ is the natural redundancy of complex normalization.

Status. Proved in this CF.

Proof. The unit circle in $\mathbb{C}$ is exactly $U(1) = \{e^{i\Lambda} : \Lambda \in \mathbb{R}\}$. Acting on a normalized fiber element by $e^{i\Lambda}$ preserves normalization and changes only phase, not the ray in projective Hilbert space. The local version, where $\Lambda = \Lambda(x)$ is a C^1 function on $\mathcal{M}$, is forced by compatibility with the C^1 structure of the representative section. This is not an additional postulate. It is the natural consequence of carrying a complex normalized state on a C^1 carrier. $\square$

2.6 What is and is not already present in $\mathcal{M}$

The derivation above does **not** say that all spacetime-bearing structure is primitive in $\mathcal{M}$. It says something more precise.

Derived and then inherited at the CF00 stage in $\mathcal{M}$: - carrier-domain locality (from Theorem 2.5.3), - local differential structure (from Theorem 2.5.4), - carrier-domain distance $d_{\mathcal{M}}$ (from the $\mathbb{R}$ completion metric of Theorem 2.5.1), - and the minimal carrier topology needed for local representative variation.

Not primitive in $\mathcal{M}$: - quotient geometry, - metric and curvature as physical geometry, - the reversible/irreversible dynamical split, - locality-bearing support, - support neighborhoods, - overlap-based gauge structures, - bond or link variables, - holonomy observables.

Thus $\mathcal{M}$ is inherited at the CF00 stage as a derived carrier input, not as already-formed physical spacetime metric or connection structure. Those later structures are induced or derived from the representative ontology built on $\mathcal{M}$.

2.7 Falsification path for stage inheritance

CF00 would fail as the correct bridge stage if any theorem in this document truly required one of the following as an additional primitive insertion beyond the inherited post-2D stack and the structures derived from it:

- a primitive support or adjacency graph,
- a primitive constitutive bond or link structure,
- a primitive gauge connection or Wilson data,
- a primitive metric or curvature tensor,

- or a primitive engine law not induced from quotient geometry.

If any such insertion were necessary, then CF00 would no longer be the correct induced-geometry bridge between CF000 and CF01.

3. Primitive Ontology and Representative Structure

3.1 Carrier domain

The carrier domain $\mathcal{M}$ is the connected C^1 domain derived in Theorem 2.5.5 from the CF000 invariant stack. Its role at this stage is inherited input in the sense of Theorem 2.4.1: earned, not assumed.

No primitive support relation on $\mathcal{M}$ is assumed. No adjacency graph, neighbor graph, bond graph, or locality-bearing edge set is part of the ontology carried here.

The only spatial structure used at stage entry is:

1. the derived local differential structure of $\mathcal{M}$,
2. and, when a cone statement is later made, a carrier-domain distance $d_{\mathcal{M}}$ used only to measure that cone.

The distance $d_{\mathcal{M}}$ does not define support. It only measures the region in which a support relation generated by the law must lie.

3.2 Effective Hilbert-bundle language

By Theorem 2.5.6, the effective complex Hilbert bundle

$$\pi : \mathcal{H} \rightarrow \mathcal{M}$$

is the derived fiber structure over $\mathcal{M}$ carrying the representative state-family. The word *effective* is used literally: each fiber $\mathcal{H}_x$ carries only the minimum inner-product structure needed to define normalization, overlap, projected variation, and local phase redundancy.

This is carrier language. It is not a claim that the effective Hilbert bundle is the final micro-ontology of the universe.

3.3 Representative state-family

A representative section is a C^1 map

$$\psi : \mathcal{M} \rightarrow \mathbb{S}(\mathcal{H}), \quad x \mapsto |\psi(x)\rangle,$$

with fiberwise normalization

$$\langle \psi(x), \psi(x) \rangle_x = 1 \quad \text{for all } x \in \mathcal{M}.$$

The state at this branch level is therefore:

- the local normalized representative state-family,
- together with its admissible local variation class.

3.4 Local $U(1)$ redundancy

The representative is redundant under local phase:

$$|\psi(x)\rangle \sim e^{i\Lambda(x)} |\psi(x)\rangle, \quad \Lambda \in C^1(\mathcal{M}, \mathbb{R}).$$

This redundancy is derived in Theorem 2.5.7 from the complex phase structure of the Hilbert bundle. It is not an additional postulate. The physical object is the quotient-physical content that survives removal of the self-parallel phase direction.

3.5 Derived representative theorem

Theorem 3.5.1 (Derived representative ontology).

The representative state-family on $\mathcal{M}$ together with local $U(1)$ redundancy are the first derived objects of this stage, earned from the CF000 invariant stack via Theorems 2.5.1–2.5.7. Support, adjacency, gauge connection, holonomy, bond structure, observable structure, and spacetime-bearing locality relations remain forbidden as primitive insertions.

Status. Proved in this CF.

Proof. The data listed above are exactly the data needed to define local representative variation and representative redundancy once the carrier, number fields, topology, differentiability, and Hilbert-bundle structure have been earned. Any stronger relational structure such as support graphs, holonomies, constitutive bonds, or gauge link variables would add relational content not yet earned from the representative ontology. Because the purpose of CF00 is to derive those structures, they cannot be inserted primitively without circularity. Therefore the first derived ontology of the stage is exhausted by the representative state-family on $\mathcal{M}$, the fiberwise inner-product structure, and the derived local $U(1)$ redundancy. $\square$

3.6 Falsification path for primitive ontology

The representative ontology theorem fails if any downstream derivation truly requires a primitive insertion of:

- support or adjacency,
- constitutive bond/link structure,
- gauge connection or Wilson data,
- or metric/curvature geometry.

If such a primitive insertion were necessary, the derived representative ontology would be too shallow.

4. Quotient-Physical Local Variation

4.1 Vertical and horizontal projectors

Define the rank-one projector onto the representative line by

$$P_{\parallel} := |\psi\rangle\langle\psi|,$$

and the complementary projector by

$$P_{\perp} := 1 - |\psi\rangle\langle\psi|.$$

For any admissible local variation v of the representative field,

$$v = v_{\parallel} + v_{\perp}, \quad v_{\parallel} = P_{\parallel}v, \quad v_{\perp} = P_{\perp}v.$$

4.2 Why the self-parallel direction is redundant

Under local rephasing

$$|\psi\rangle \mapsto e^{i\Lambda}|\psi\rangle,$$

the infinitesimal variation generated by changing Λ is proportional to $i|\psi\rangle$. This lies in the one-dimensional line spanned by $|\psi\rangle$. Therefore that direction changes only the representative chart, not the ray state. It is the local self-parallel phase direction.

4.3 Quotient-physical local variation theorem

Theorem 4.3.1 (Quotient-physical local variation).

The physically meaningful local variation is the horizontal variation class defined by $P_{\perp}$. The self-parallel direction defined by $P_{\parallel}$ is redundant and must be quotiented out before any physical geometric or dynamical object is constructed.

Proof.

The representative redundancy is exactly the local phase freedom. Infinitesimal variation in that phase direction lies in the representative line and therefore changes only the representative, not the ray state. Any local bilinear or dynamical construction that is meant to represent physical variation must therefore remove the $P_{\parallel}$ component. The surviving local content is exactly the horizontal class $P_{\perp}v$. $\square$

4.4 Physical state space at this layer

The physical local state at this layer is therefore not $|\psi\rangle$ itself but the quotient class of representatives under local phase. The projected local variation lives naturally on this quotient state space.

4.5 Falsification path for quotient construction

The quotient theorem fails if:

- the vertical direction contributes to any gauge-invariant local bilinear object,
- or the local phase direction produces distinct physical local predictions after quotienting.

Either failure would show that the phase direction is not truly redundant.

5. Induced Geometry from Projected Variation

5.1 Projected local derivatives

Let x^μ be local coordinates on $\mathcal{M}$. The raw derivative $\partial_\mu|\psi\rangle$ contains both physical and pure-phase content. Define the projected derivative by

$$|\partial_\mu\psi\rangle_\perp := P_\perp|\partial_\mu\psi\rangle = |\partial_\mu\psi\rangle - |\psi\rangle\langle\psi|\partial_\mu\psi\rangle.$$

This is the quotient-physical local derivative.

5.2 Definition of the quantum geometric tensor

Define

$$Q_{\mu\nu} = \langle\partial_\mu\psi | \partial_\nu\psi\rangle - \langle\partial_\mu\psi | \psi\rangle\langle\psi | \partial_\nu\psi\rangle.$$

Equivalently,

$$Q_{\mu\nu} = \langle\partial_\mu\psi | P_\perp\partial_\nu\psi\rangle.$$

Thus $Q_{\mu\nu}$ depends only on projected variation.

5.3 Gauge invariance of the QGT

Under local rephasing,

$$|\psi\rangle \mapsto e^{i\Lambda}|\psi\rangle,$$

one has

$$|\partial_\mu\psi\rangle \mapsto e^{i\Lambda}(|\partial_\mu\psi\rangle + i(\partial_\mu\Lambda)|\psi\rangle).$$

The added term is vertical because it is proportional to $|\psi\rangle$. Applying $P_\perp$ removes it. Therefore the projected derivative is unchanged up to the common phase factor. The Hermitian bilinear form built from projected derivatives is therefore gauge-invariant.

5.4 Induced-QGT theorem

Theorem 5.4.1 (Induced quantum geometric tensor).

The tensor $Q_{\mu\nu}$ is the first gauge-invariant bilinear geometric object induced by the primitive ontology. It is induced from quotient-physical local variation and is not primitive.

Proof.

The primitive ontology supplies only local representative states and their quotient-physical local variation. The projected derivative is the first local object that survives the quotient. The Hermitian bilinear form built from those projected derivatives is exactly $Q_{\mu\nu}$. Since it is gauge-invariant and built solely from quotient-physical local variation, it is the first induced geometric object. It cannot be inserted primitively without violating the primitive ontology theorem. $\square$

5.5 Metric and curvature split

Define

$$g_{\mu\nu} := \operatorname{Re} Q_{\mu\nu}, \quad \Omega_{\mu\nu} := -2 \operatorname{Im} Q_{\mu\nu}.$$

Then

$$Q_{\mu\nu} = g_{\mu\nu} - \frac{i}{2} \Omega_{\mu\nu}.$$

The tensor g is real and symmetric. The tensor Ω is real and antisymmetric.

5.6 Induced metric/curvature theorem

Theorem 5.6.1 (Induced metric and curvature sectors).

The metric sector g and the curvature sector Ω are induced from the QGT and therefore from quotient-physical local variation. Neither is primitive.

Proof.

Every Hermitian bilinear form decomposes uniquely into its real symmetric part and its imaginary antisymmetric part. Applying that decomposition to the induced QGT defines g and Ω . Since Q itself is induced, gauge-invariant, and quotient-physical, the same is true of its metric and curvature sectors. $\square$

5.7 Falsification path for induced geometry

The induced-geometry chain fails if:

- the projected derivative is not gauge-invariant,
- the QGT depends on vertical representative content,
- the split into g and Ω is not uniquely induced from Q ,
- or later dynamics require geometric objects not earned from the induced QGT.

6. Tangent Decomposition and Emergent Dynamical Engine

6.1 Tangent decomposition of representative evolution

Let $t \mapsto \psi(t)$ be an admissible representative trajectory. Since

$$\langle \psi, \psi \rangle = 1,$$

differentiate in time:

$$\frac{d}{dt} \langle \psi, \psi \rangle = \langle \dot{\psi}, \psi \rangle + \langle \psi, \dot{\psi} \rangle = 2 \operatorname{Re} \langle \psi, \dot{\psi} \rangle = 0.$$

Hence $\langle \psi, \dot{\psi} \rangle$ is purely imaginary. Therefore there exists a real scalar functional $\alpha[\psi]$ such that

$$\langle \psi, \dot{\psi} \rangle = -i \alpha[\psi].$$

Define

$$\chi[\psi] := \dot{\psi} + i \alpha[\psi] \psi.$$

Then

$$\langle \psi, \chi[\psi] \rangle = \langle \psi, \dot{\psi} \rangle + i \alpha[\psi] \langle \psi, \psi \rangle = -i \alpha[\psi] + i \alpha[\psi] = 0.$$

Therefore every norm-preserving representative evolution decomposes uniquely as

$$\dot{\psi} = -i \alpha[\psi] \psi + \chi[\psi], \quad \langle \psi, \chi[\psi] \rangle = 0.$$

6.2 Interpretation of the decomposition

The term $-i \alpha[\psi] \psi$ is vertical. It is pure representative phase motion. The term $\chi[\psi]$ is horizontal. It is the quotient-physical evolution class.

6.3 The geometric operators available for dynamics

Once the quotient geometry has been induced, there are exactly two geometric structures available at this level:

- the antisymmetric sector Ω ,
- the symmetric sector g .

On the regular quotient sectors these define bundle maps from cotangent data to tangent data:

$$J := \Omega^\sharp, \quad M := g^\sharp.$$

This is the first appearance of plain M in the document. It is the dissipative symmetric operator induced from the quotient geometry. It is not the carrier domain.

6.4 Why the mixed law is forced

A local quotient-physical law built only from induced geometry and scalar generators must use only the maps already earned from the induced geometry. At this level, those are precisely J and M .

The antisymmetric operator J generates the reversible sector because it acts on scalar differentials through the curvature structure. The symmetric operator M generates the irreversible sector because it acts on scalar differentials through the metric structure. If both sectors are physically present and no extra primitive dynamical tensor may be inserted, then the most general scalar-generated local law is their sum.

Connection to the primitive bifurcation invariant. The above geometric argument shows the mixed law is forced by the induced quotient geometry of the carrier. At a deeper level, the reason both sectors must appear is the primitive bifurcation invariant itself (A(-1)): the invariant cannot discharge into the J -pole alone (which would give a fully conservative, reversible system with no entropy production — a collapse toward the pole of absolute order) and cannot discharge into the M -pole alone (which would give a fully dissipative system with no conserved structure — a collapse toward the pole of absolute disorder). The $J \oplus M$ split is therefore not merely a geometric consequence at this layer; it is the expression of the primitive two-pole opposition at the dynamical layer. Energy (J -limb) and entropy (M -limb) are the names the two irreconcilable poles take on once a carrier and dynamics exist.

6.5 Emergent engine theorem

Theorem 6.5.1 (Emergent scalar-generated mixed dynamical architecture).

Let the quotient geometry be given by the induced pair (g, Ω) . Suppose the horizontal law is required to satisfy all of the following:

1. it is local on $\mathcal{M}$,
2. it acts only on quotient-physical data,
3. it is generated by scalar state functionals,
4. it contains both the reversible sector carried by the antisymmetric induced geometry and the irreversible sector carried by the symmetric induced geometry,
5. it does not insert any further primitive generator tensor not already earned from the induced geometry.

Then the horizontal law must take the scalar-generated mixed form

$$\bar{\chi} = J d\mathcal{I} + M d\Sigma = \Omega^\sharp d\mathcal{I} + g^\sharp d\Sigma,$$

for scalar generators $\mathcal{I}$ and Σ .

Proof.

Condition (2) excludes vertical representative content. Condition (5) excludes any new primitive tensor beyond the induced geometric data. Thus the only available local bundle maps from scalar differentials to quotient tangent vectors are those induced from Ω and g .

An antisymmetric bilinear structure acting on a scalar differential gives the reversible local transport allowed by the induced curvature sector. A symmetric bilinear structure acting on a scalar differential gives the gradient-like local transport allowed by the induced metric sector. If both sectors must appear and both must be scalar-generated, then every admissible horizontal law is the sum of one term generated by the antisymmetric operator and one term generated

by the symmetric operator. No third independent primitive sector may appear by condition (5). Therefore the general scalar-generated law has the form

$$\bar{\chi} = J d\mathcal{I} + M d\Sigma.$$

Since $J = \Omega^\sharp$ and $M = g^\sharp$ on the regular quotient sectors, this is exactly

$$\bar{\chi} = \Omega^\sharp d\mathcal{I} + g^\sharp d\Sigma.$$

Thus the dynamical engine is not imported. It is forced by the induced quotient geometry under the stated conditions. $\square$

6.6 Honest strength statement

The theorem above does not claim that no deeper pre-geometric dynamics could ever exist in some other branch. It claims that, once the primitive representative ontology of this branch is accepted, the law-bearing engine at this level is forced into the scalar-generated mixed architecture just derived.

6.7 Falsification path for emergent engine

The emergent-engine theorem fails if any of the following occur:

- a primitive J -sector or M -sector is needed,
- a third primitive dynamical sector is needed,
- or the induced geometry cannot generate both reversible and irreversible channels.

7. Constitutive Narrowing and Reservoir Equivalence

7.1 Need for a constitutive sector

The mixed law from Section 6 is structural. To obtain a concrete physical engine one must specify the invariant generator $\mathcal{I}$ and entropy generator Σ .

7.2 Minimal conservative/entropy-bearing form

The narrowest admissible constitutive form at this level is

$$\mathcal{I}[\psi, u] = \int_{\mathcal{M}} \left(\frac{\tau}{2} |D_t \psi_\perp|_g^2 + \frac{D}{2} \sum_{i=1}^d |D_i \psi_\perp|_g^2 + V([\psi]) + u \right) d\mu,$$

and

$$\Sigma[\psi, u] = \int_{\mathcal{M}} s(u) d\mu,$$

with

$$s \in C^2, \quad s'(u) > 0, \quad s''(u) < 0.$$

Here:

- $V([\psi])$ is a local gauge-invariant quotient potential,
- τ and D determine the telegraph/Cattaneo principal part,
- u is the local entropy-bearing reservoir variable,
- and $d\mu$ is the carrier-domain integration measure.

7.3 Why the reservoir is required

The irreversible sector cannot be closed on the conservative variables alone without collapsing the distinction between the invariant generator and the entropy generator. The reservoir variable is therefore not ornamental. It is the minimal additional scalar field needed to carry the irreversible channel while preserving that distinction.

7.4 Reservoir-coordinate conjugacy lemma

Let

$$\mathcal{I}[\psi, u] = \mathcal{I}_0[\psi] + \int_{\mathcal{M}} u \, d\mu, \quad \Sigma[\psi, u] = \int_{\mathcal{M}} s(u) \, d\mu,$$

where $\mathcal{I}_0[\psi]$ is the conservative sector and $s'(u) > 0$ on the reservoir range. Define

$$\Phi : (\psi, u) \mapsto (\psi, \sigma), \quad \sigma = s(u).$$

Lemma 7.4.1 (Reservoir-coordinate conjugacy).

On every region where $s'(u) > 0$, the map Φ is a local C^2 diffeomorphism on the reservoir sector. In the transformed coordinates one has

$$\tilde{\mathcal{I}}[\psi, \sigma] = \mathcal{I}_0[\psi] + \int_{\mathcal{M}} s^{-1}(\sigma) \, d\mu, \quad \tilde{\Sigma}[\psi, \sigma] = \int_{\mathcal{M}} \sigma \, d\mu.$$

The primitive law in (ψ, u) and the primitive law in (ψ, σ) are locally conjugate. Their degeneracy content, entropy-production sign, derived support relation, and runtime-admissibility content are the same physical content written in different reservoir charts.

Proof.

Because $s'(u) > 0$, the inverse function theorem gives a local inverse $u = s^{-1}(\sigma)$, so Φ is a local diffeomorphism.

Let X be the primitive law vector field in (ψ, u) coordinates. The transformed law is

$$\tilde{X} = T\Phi X \Phi^{-1}.$$

Thus the two laws are dynamically conjugate.

The degeneracy statements are kernel statements for the induced symmetric and antisymmetric operators acting on the differentials of the scalar generators. Since those tensors and differentials transform covariantly under Φ , the vanishing or nonvanishing of those kernel conditions is unchanged. Therefore the degeneracy content is preserved.

The transformed entropy is

$$\tilde{\Sigma}[\psi, \sigma] = \Sigma[\psi, s^{-1}(\sigma)].$$

Hence along corresponding trajectories one has

$$\frac{d}{dt} \tilde{\Sigma} = \frac{d}{dt} \Sigma.$$

So the sign of entropy production is unchanged.

If U_{t,t_0} is the nonlinear solution operator, then

$$\tilde{U}_{t,t_0} = \Phi \circ U_{t,t_0} \circ \Phi^{-1}.$$

Differentiating gives

$$D\tilde{U}_{t,t_0} = D\Phi \circ DU_{t,t_0} \circ D\Phi^{-1}.$$

Therefore the support relation defined by nonzero propagated horizontal perturbations is preserved. Since runtime admissibility is defined only by faithful discretization of the carrier, quotient, law, and derived support relation, the runtime-admissibility content is also preserved. $\square$

7.5 Constitutive equivalence theorem

Theorem 7.5.1 (Theorem-grade constitutive equivalence class).

The entropy-bearing constitutive sector is theorem-grade at the level of the equivalence class

$$s \in C^2, \quad s'(u) > 0, \quad s''(u) < 0,$$

modulo smooth strictly monotone reservoir-coordinate reparameterization.

Proof.

By Lemma 7.4.1, any two entropy charts related by such a reparameterization are locally conjugate, preserve the mixed law structure, preserve the degeneracy content, preserve entropy-production sign, preserve the support theorem content, and preserve runtime-admissibility content. Therefore the theorem-grade constitutive content is the equivalence class, not any one special chart. $\square$

7.6 Canonical execution specialization

A canonical execution specialization is

$$s(u) = k_B \log u, \quad u > 0.$$

This specialization belongs to the theorem-grade equivalence class but is not uniquely forced by the theorem.

7.7 Falsification path for constitutive narrowing

The constitutive equivalence theorem fails if smooth strictly monotone reservoir-coordinate changes alter:

- the mixed-law structure,
- degeneracy content,
- the sign of entropy production,
- the derived support relation,
- or the runtime-admissibility content.

8. Derived Locality, Support, and Gauge-Hosting

8.1 Nonlinear solution operator

Let Z denote the primitive state space of admissible fields (ψ, u) modulo local $U(1)$ redundancy on the representative sector. Let

$$U_{t,t_0} : Z \rightarrow Z$$

denote the nonlinear solution operator of the closed primitive law on its interval of existence.

For each admissible trajectory $z(\cdot)$ and each pair $t \geq t_0$, let

$$DU_{t,t_0} \big|_{z(t_0)}$$

denote the Fréchet derivative of the solution operator at the initial state.

8.2 Derived support relation

A horizontal perturbation δz_0 at time t_0 is localized near $x_0 \in \mathcal{M}$ if its support lies in an arbitrarily small neighborhood of x_0 in the primitive carrier domain.

Define

$$(x_0, t_0) \rightsquigarrow (x, t)$$

if and only if there exists a localized horizontal perturbation δz_0 near x_0 such that

$$(DU_{t,t_0} \big|_{z(t_0)} \delta z_0)(x) \neq 0.$$

Define the exact support set by

$$\text{Supp}(t; t_0, x_0) = \overline{\{x \in \mathcal{M} : (x_0, t_0) \rightsquigarrow (x, t)\}}.$$

8.3 Derived-support theorem

Theorem 8.3.1 (Derived support from the law).

For every admissible trajectory of the closed primitive law, the locality-bearing support relation is the support relation induced by the retarded propagation of horizontal perturbations under the Fréchet derivative of the nonlinear solution operator. It is not primitive adjacency, not thresholded observable recovery, and not graph structure inserted by hand.

If the principal part of the closed primitive law is of telegraph/Cattaneo type with effective characteristic speed c_{eff} , then

$$\text{Supp}(t; t_0, x_0) \subseteq \{x \in \mathcal{M} : d_{\mathcal{M}}(x, x_0) \leq c_{\text{eff}}(t - t_0)\}$$

for all $t \geq t_0$ within the domain of existence.

Proof.

The support relation is defined entirely through the law via its nonlinear solution operator and the propagation of horizontal perturbations under its Fréchet derivative. Therefore support is generated by the law itself.

Because the perturbations are horizontal, representative phase redundancy does not affect whether a perturbation propagates nontrivially. So the support object is quotient-physical.

If the principal part of the law is telegraph/Cattaneo type, then the domain of dependence of perturbations is bounded by the corresponding characteristic cone measured with the carrier-domain distance $d_{\mathcal{M}}$. Therefore the derived support set lies inside that cone. $\square$

8.4 Observable-independence corollary**Corollary 8.4.1 (Observable-independence).**

Let $\mathcal{F}$ be any separating family of local gauge-invariant observables on the horizontal quotient dynamics. Then $\mathcal{F}$ witnesses the derived support relation but does not define it. The support set $\text{Supp}(t; t_0, x_0)$ is independent of the separating family used to detect propagated perturbations.

Proof.

The support object is defined from the law itself through DU_{t,t_0} . Any separating witness family merely detects whether the propagated perturbation is nonzero. It does not alter the propagation. Therefore different separating witness families recover the same support object. $\square$

8.5 Non-circularity lemma**Lemma 8.5.1 (Non-circularity of derived support).**

The primitive inputs of the formalism are the carrier domain $\mathcal{M}$, the representative field, the reservoir field, and the closed primitive law on those fields. No support relation, adjacency graph, or neighborhood graph on $\mathcal{M}$ is primitive. The support relation on $\mathcal{M}$ is derived from the retarded propagation of horizontal perturbations under the law.

Proof.

The only primitive role of $\mathcal{M}$ is to provide the arena on which fields are defined. The support relation is a subset of $\mathcal{M} \times \mathcal{M} \times \mathbb{R}^2$ selected by nonzero propagation under the solution operator. Since that relation is computed from the law and is not independently specified, no primitive locality-bearing adjacency is inserted. $\square$

8.6 Local overlap cover and loop-hosting lemma

Let $\psi : \mathcal{M} \rightarrow \mathbb{S}(\mathcal{H})$ be continuous and normalized. For fixed $x \in \mathcal{M}$, the map

$$y \mapsto \langle \psi(x), \psi(y) \rangle$$

is continuous and equals 1 at $y = x$.

Lemma 8.6.1 (Local overlap cover and loop-hosting).

For every $x \in \mathcal{M}$ there exists an open neighborhood $U_x \subset \mathcal{M}$ such that

$$\langle \psi(x), \psi(y) \rangle \neq 0 \quad \text{for all } y \in U_x.$$

Hence the family $\{U_x\}_{x \in \mathcal{M}}$ is an open overlap cover on which local overlap phases

$$\mathcal{U}(x, y) = \frac{\langle \psi(x), \psi(y) \rangle}{|\langle \psi(x), \psi(y) \rangle|}$$

are well-defined whenever x and y lie in a common overlap neighborhood.

If a support-generated neighborhood $V \subset \text{Supp}(t; t_0, x_0)$ is contained in one such overlap neighborhood, then:

1. local link phases on V are well-defined and gauge-covariant;
2. any sufficiently small loop in V admits a discrete holonomy built from ordered products of local link phases;
3. plaquette and Wilson-loop constructions on such sufficiently small support-generated loops are well-defined.

Proof.

Continuity of the overlap map and normalization at coincidence imply the existence of a neighborhood U_x on which the overlap remains nonzero. The normalized overlap phase is therefore well-defined on each such neighborhood. Under local $U(1)$ rephasing, the normalized overlap phase transforms covariantly and therefore defines a local link variable.

If a support-generated neighborhood is subordinate to the overlap cover, then any sufficiently small loop in that neighborhood is covered by finitely many overlap neighborhoods. Ordered products of local link variables along the loop are therefore defined, and these products give local plaquette and Wilson-loop holonomies. $\square$

8.7 Gauge-hosting sufficiency corollary

Corollary 8.7.1 (Gauge-hosting sufficiency of derived support).

Within the scope of CF00, the derived support neighborhoods furnished by the law are sufficient to host the downstream local overlap-based connection, plaquette, and Wilson-loop machinery. Therefore gauge-holonomy structure in this branch depends on derived support and continuous representative geometry, not on inserted primitive adjacency.

Proof.

By Lemma 8.6.1, each point admits an overlap neighborhood with well-defined local link phases. By Theorem 8.3.1, the law furnishes local support neighborhoods. On sufficiently fine local scales the support neighborhoods may be taken subordinate to the overlap cover. Therefore the local overlap-based gauge machinery is structurally supported on derived support. $\square$

8.8 Falsification path for derived locality and gauge hosting

The support theorem or its corollaries fail if:

- support cannot be defined directly from the solution operator and its Fréchet derivative,
- different separating witness families define inequivalent support objects,
- support requires primitive adjacency,
- support violates the carrier-domain cone bound,
- or support-generated neighborhoods do not admit local overlap-based gauge constructions.

9. Runtime Admissibility

A runtime architecture is admissible relative to CF00 on a finite validation horizon if it satisfies the following sufficient conditions.

Theorem 9.1 (Runtime admissibility).

Let $\mathcal{R}_h$ be a discretized implementation. Suppose the following hold on a finite validation horizon.

1. $\mathcal{R}_h$ discretizes the primitive carrier variables (ψ, u) on $\mathcal{M}$ and preserves normalization and the local $U(1)$ quotient structure up to bounded residuals.
2. $\mathcal{R}_h$ discretizes the closed scalar-generated primitive law together with the theorem-grade constitutive equivalence class of the entropy-bearing reservoir sector.
3. $\mathcal{R}_h$ computes support from the discrete retarded propagator of the discretized law and does not insert adjacency, support, or neighborhood graphs as primitive ontological inputs.
4. $\mathcal{R}_h$ preserves, up to bounded residuals on the validation horizon,
 - induced-QGT regularity,
 - the metric/curvature split,
 - the mixed generator structure,
 - entropy-production sign,
 - cone inclusion of derived support relative to $d_{\mathcal{M}}$,
 - and the overlap-cover conditions needed for local overlap-based gauge constructions.
5. $\mathcal{R}_h$ introduces no undeclared conserved quantities, primitive support structures, or gauge inconsistencies beyond those allowed by the formalism.

Then $\mathcal{R}_h$ is an admissible runtime realization of CF00 on that validation horizon.

Proof.

Sections 2 through 7 determine the ontologically load-bearing structures of this branch: primitive representative carrier, quotient structure, induced geometry, scalar-generated mixed law, constitutive equivalence class, derived support relation, and local gauge-hosting sufficiency. A runtime satisfying conditions (1) through (5) preserves exactly those

structures, up to bounded residuals on the stated horizon, and does not insert structures forbidden by the primitive ontology theorem. Therefore it is an admissible realization of CF00 on that horizon. $\square$

This is a theorem of admissibility. It is not a necessity claim, not an “unlock” slogan, and not a statement that every concrete implementation automatically satisfies its hypotheses.

9.1 Falsification path for runtime admissibility

A concrete implementation fails the admissibility theorem if it:

- inserts primitive adjacency,
- fails to preserve the quotient or induced-geometry structure,
- violates the carrier-domain cone bound,
- or introduces hidden integrals or primitive support structures not licensed by the formalism.

The executable notebook of CF00 may compute and visualize these checks, but it does not supply missing proof.

10. Validation and Falsification Inside the Formalism

Validation belongs to the formalism itself as falsification logic. It is not an external sink.

10.1 Primitive carrier falsifiers

The primitive carrier theorem fails if the formalism requires any primitive insertion of:

- adjacency or support graph on $\mathcal{M}$,
- holonomy or gauge data before quotient-physical variation is derived,
- constitutive bond structure before the mixed law is derived,
- or observable structure before support is derived from the law.

10.2 Induced geometry falsifiers

The induced-geometry chain fails if:

- projected variation is not gauge-invariant,
- the QGT depends on vertical representative content,
- the split into metric and curvature is not uniquely induced from the QGT,
- or later dynamics require geometric objects not earned from the induced QGT.

10.3 Mixed-law falsifiers

The emergent-engine theorem fails if:

- a primitive J -sector or M -sector is needed,
- a third primitive dynamical sector is needed,
- or the induced geometry cannot generate both reversible and irreversible channels.

10.4 Constitutive equivalence falsifiers

The constitutive equivalence theorem fails if reservoir-coordinate changes alter:

- the mixed-law structure,
- degeneracy content,
- the sign of entropy production,
- the derived support relation,
- or the runtime-admissibility content.

10.5 Derived support falsifiers

The support theorem fails if:

- support cannot be defined directly from the nonlinear solution operator and its Fréchet derivative,
- observables define different support objects rather than witnessing one,
- support requires primitive adjacency,
- support violates the $d_{\mathcal{M}}$ cone bound,
- or support-generated neighborhoods fail to admit local overlap-based gauge constructions.

10.6 Runtime falsifiers

A runtime realization fails admissibility if it:

- inserts primitive adjacency,
- fails to preserve the quotient or induced-geometry structure,
- violates the carrier-domain cone bound,
- or introduces hidden integrals or primitive support structures not licensed by the formalism.

11. Downstream Placement in the Canon

CF00 is the current bedrock formalism for this branch. It is physically first, even though it is historically later.

11.1 CF01

CF01 is reclassified as a downstream effective engine formalism. It remains historically important because it was the first successful explicit engine-layer document. It is not load-bearing for the root derivation carried here.

11.2 CF11

CF11 is reclassified as a downstream derived-limit module. Its scalar-generated law form and entropy-bearing reservoir logic are consistent with CF00, but they do not substitute for root derivation once CF00 exists.

11.3 Other CFs

Other cited CFs remain relevant as downstream modules, consistency witnesses, or provenance records. They do not carry root proof burden for this branch once CF00 is in place.

12. Internal Canon References

The following internal canon modules are retained as provenance, downstream placement, or consistency witnesses. They are not load-bearing substitutes for the derivation carried here.

- **CF01** — provenance and downstream effective engine placement.
- **CF02** — provenance and downstream contact/Generic compatibility witness.
- **CF04** — provenance and downstream finite-speed / cone witness.
- **CF05** — provenance and downstream closure / no-hidden-integrals witness.
- **CF06** — provenance and downstream Fisher / information-geometric witness.
- **CF09** — provenance and downstream overlap/holonomy witness.
- **CF11** — provenance and downstream derived-limit reservoir/entropy witness.

Where this document reproduces, absorbs, or strengthens structures historically associated with those documents, the proof burden is carried here and not delegated back to them.

13. External Mathematical Background

The following external mathematical categories provide language and sharpening tools. They are not canon authorities.

1. **Projective Hilbert space and geometric quantum mechanics**
Used for representative-to-ray quotient language, vertical/horizontal decomposition, and local overlap geometry.
2. **Quantum geometric tensor, Fubini–Study, and Berry geometry**
Used for the gauge-invariant Hermitian bilinear form and the induced split into metric and curvature sectors.
3. **GENERIC, contact, and metriplectic structure**
Used for reversible/irreversible mixed-generator language and entropy-production form.
4. **Hyperbolic / telegraph propagation and differentiable nonlinear flow operators**
Used for the support theorem through the nonlinear solution operator and its Fréchet derivative.

14. Novel Contributions of CF00

The following are newly formalized here as root-level results.

1. The integration of the derived carrier, quotient-physical variation, induced QGT, and induced metric/curvature sectors into a single dependency-clean derivation.
2. The derivation of the scalar-generated mixed dynamical architecture from induced geometry rather than importing an engine from downstream canon.
3. The theorem-grade constitutive equivalence class under reservoir-coordinate conjugacy.
4. The support theorem formulated directly from the nonlinear solution operator and its Fréchet derivative.
5. The explicit non-circularity theorem for derived support.
6. The local overlap-cover and loop-hosting lemma proving gauge-hosting sufficiency of derived support neighborhoods.
7. The runtime admissibility theorem stated in the strongest honest form that follows from the formalism.
8. The reclassification of CF01 and CF11 as downstream documents relative to this root formalism.
9. The stage-status clarification of the carrier domain $\mathcal{M}$ within the present branch.
10. The derivation of $\mathbb{R}$, $\mathbb{C}$, topology, the primitive half-turn constant π of the continuous ORS orbit, C^1 differentiability, the carrier domain $\mathcal{M}$, the effective Hilbert bundle, and local $U(1)$ redundancy from the CF000 post-2D invariant stack, eliminating all primitive assumptions about the carrier structure.

15. Explicit Non-Claims

CF00 does not claim the following.

1. It does not claim that the effective Hilbert-bundle language is final micro-ontology.
2. It does not claim necessity in the runtime admissibility theorem.
3. It does not claim a unique entropy chart; it proves only the equivalence class under reservoir-coordinate conjugacy.
4. It does not claim that every downstream extension of VDM is exhausted by this branch.

CF00 also does not claim that no deeper theory could ever exist. What it does claim is stronger and more precise: no deeper substrate beneath $\mathcal{M}$ is mathematically forced within the current stage once the post-2D invariant stack has been used to derive the carrier. Therefore CF00 is stage-complete for this branch unless and until a theorem proves otherwise.

16. Stage Status of the Carrier Domain $\mathcal{M}$

The question answered here is not whether deeper metaphysics can be imagined. The question is whether the formalism, at its present stage, forces anything deeper than the carrier domain already derived in §2.5.

16.1 What $\mathcal{M}$ is in CF00

$\mathcal{M}$ is the derived carrier arena of the branch. It is the connected C^1 domain on which the representative state-family and reservoir field live.

16.2 What $\mathcal{M}$ is not in CF00

$\mathcal{M}$ is not:

- a primitive metric geometry,
- a primitive gauge connection,
- a primitive support graph,
- a primitive bond structure,
- or the mixed dynamical engine.

16.3 What is already encoded in $\mathcal{M}$

What is already encoded in $\mathcal{M}$ at this stage is only what has been derived before use: coordinate-bearing real completion, complex fiber compatibility, topology, C^1 variation structure, and the minimal locality needed to state local variation and cone bounds.

16.4 Stage-status statement

Within CF00, $\mathcal{M}$ is the inherited derived input — derived in §2.5 from the CF000 invariant stack, and then serving as the carrier arena from which every later load-bearing structure in this document is induced. No primitive insertion of $\mathbb{C}$, topology, differentiability, or the carrier itself is made. All are earned before being used.

16.5 Falsifier for stage status

The claim that $\mathcal{M}$ is the correct carrier stage for CF00 is falsified if any theorem requires:

- a deeper substrate to define representative redundancy,
- a deeper substrate to define projected variation,
- a deeper substrate to derive the QGT,
- a deeper substrate to derive the mixed law,
- or a deeper substrate to derive support.

No such requirement is proved in this branch. Therefore the stage-status claim stands.

17. CF Status Note

- **Theorem-grade content carried here:** the derivation of the carrier from the post-2D invariant stack, derived representative ontology, quotient-physical local variation, induced QGT, induced metric/curvature split, tangent decomposition, scalar-generated mixed law, constitutive equivalence under reservoir-coordinate conjugacy, derived support from the nonlinear solution operator and its Fréchet derivative, observable-independence, non-circularity, local overlap-cover / loop-hosting, gauge-hosting sufficiency, and runtime admissibility.
- **What remains downstream rather than missing:** executable realization, numerical witness generation, figure production, bounded-residual checks for concrete implementations, and the explicit QGT-to-metriplectic bracket mapping developed in CF01.
- **What is not left in suspense:** whether the carrier can still be primitively inserted without derivation. It cannot; that burden has now been moved upstream and closed in the present paper.

- **Completion verdict:** CF00 is complete as the current carrier-derivation and induced-geometry formalism for this branch of VDM physics.